

Graph-Based Retrieval-Augmented Generation for Multi-hop Question Answering: A Two-Arm Evaluation of Four Frameworks

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Graph-Based Retrieval-Augmented Generation for Multi-hop Question Answering: A Two-Arm Evaluation of Four Frameworks

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Abstract

Multi-hop question answering breaks the premise of flat RAG. The passage that contains the answer only becomes recognizable after another one is read. Frameworks that structure the index as a knowledge graph respond to this problem and report gains over dense baselines. In the published evaluations, however, the same system almost always retrieves and answers, and the result does not separate what the index found from what the pipeline did with the evidence. This study evaluates LightRAG, Microsoft GraphRAG, HippoRAG 2 and Cogne in two arms over the same index. In the first arm, the framework only retrieves, and a fixed reader answers. In the second, the framework answers with its native pipeline. The matrix was run over 1,000 MuSiQue questions and replicated on 1,000 questions of 2WikiMultiHopQA, with an LLM judge validated by human labeling and paired McNemar tests under Holm correction. With the fixed reader, the effect varies by framework. LightRAG is 5.7 and 5.3 points of strict accuracy above the dense baseline, while Cogne is 6.2 and 6.3 points below. Since the systems do not deliver the same amount of context, these differences do not isolate the effect of the graph. On 2WikiMultiHopQA, the four native pipelines score below their controlled counterparts. The difference ranges from 7.7 to 17.6 points, with refusal rates between 16.0% and 26.0%. The decomposition by question type helps to interpret the difference between the datasets, although the analysis is descriptive.

Keywords: retrieval-augmented generation, knowledge graphs, multi-hop question answering, LLM as a judge, evaluation.

1 Introduction

Retrieval-augmented generation (RAG) grounds the answers of a language model in documents retrieved at query time, instead of relying on what the model memorized during training (Lewis et al., 2020). In its dominant form, the corpus is embedded in a vector space and the retriever returns the passages closest to the question (Karpukhin et al., 2020). This works when a single passage contains the answer. Multi-hop questions break that premise. To answer “in which country was the director of this film born?”, the passage that names the country can only be recognized as relevant *after* the director has been identified in another passage. Benchmarks such as MuSiQue and 2WikiMultiHopQA are built to require this composition of steps, although to different degrees.

MuSiQue filters out the compositions that a model solves without the intermediate hop, while on 2WikiMultiHopQA a model that reads one paragraph at a time still answers a good share of the questions (Trivedi et al., 2022). Dense retrievers score poorly on these sets because they order passages by similarity to the question, without considering the place they occupy in a chain of evidence (Gutiérrez et al., 2025).

One family of frameworks responds to this by structuring the index as a knowledge graph. LightRAG (Guo et al., 2025), HippoRAG 2 (Gutiérrez et al., 2025), Microsoft GraphRAG (Microsoft, 2026) and Cogneer (Topoteretes, 2026) differ in what they extract from the collection and in how they traverse the graph, and Section 2.2 describes each one.

The evaluations published by each of them are end-to-end and do not allow retrieval to be separated from the rest of the pipeline. Independent comparisons have already addressed part of this problem by adopting a common reader and the same nominal `top_k` (Zhou et al., 2025). None of them, however, covers the four systems under the two conditions (Section 2.5).

Faced with this gap, this dissertation asks two questions:

Primary. *How do four graph-based RAG frameworks compare on multi-hop question answering when the answer is produced by a controlled reader or by the native pipeline of each system?*

Secondary. *How do the context budget and the composition of the questions help in understanding the differences in performance, and what does abstention reveal about them?*

The second question is treated descriptively, within what the data allow. In this design, each framework is run twice over the same index, once in each experimental arm. In the *native* arm, the framework runs its default pipeline. In the *controlled* one, the framework only retrieves. The documents travel through the adapter with their identification preserved, and the same reader answers in every cell of that arm (Section 3.1).

Although the index is shared, the context delivered to the reader varies across the cells. The nominal retrieval parameter applies to different units in each implementation, and the requested value also differs, so that the budgets are not equivalent (Section 3.4). The matrix brings together 1,000 questions per dataset. For each dataset, the *closed-book* floor, the dense baseline and the *oracle* ceiling serve as a reference for interpreting the results of the four frameworks. An LLM judge scores each answer in two passes, and the agreement between the two is measured. The differences are tested question by question with exact McNemar under Holm correction (McNemar, 1947; Holm, 1979).

With the fixed reader, the same pattern appears on both datasets, with LightRAG and HippoRAG 2 above dense retrieval and Cogneer below. Only for Microsoft GraphRAG does the difference with respect to the baseline depend on the dataset (Section 6). Since the budgets are not equivalent, these deltas characterize the retrieval adapters, not the isolated effect of the graph structure.

Under the native pipelines, the effect is mixed on MuSiQue and negative for the four frameworks on 2WikiMultiHopQA. On that dataset, all of them score significantly below their respective controlled arms, and the pipelines refuse between one in six and one in four questions (Table 4). Part of the error that appears as an incorrect answer in the controlled arm comes to appear as a refusal in the native one. Making this difference between the two arms visible is the main contribution of this work.

The remainder of this dissertation is organized as follows. Section 2 reviews the literature on RAG. Section 3 describes the method used to compare the four frameworks. Section 4 shows how the two-arm experiment was structured, and Section 5 details its execution. Section 6 presents and discusses the results obtained on MuSiQue and 2WikiMultiHopQA. Finally, Section 7 answers the research questions and discusses the limitations of the study.

2 Related Work

Multi-hop question answering (QA) defines the problem of this study, the graph-based RAG frameworks are the object evaluated, and the instruments come from the evaluation of RAG systems. A fourth literature, that of attribution and abstention, gives a name to what is measured when the retrieved evidence is not enough. This review classifies the methods along three axes: the *index structure* (what is built at indexing time), the *retrieval operation* (how the evidence is located at query time) and the *coupling between retrieval and generation* (how much of the final result depends on each system’s own answering pipeline). The first two axes describe the substrate. The third is the reason for the two-arm design of this study.

2.1 Retrieval-Augmented Generation and Multi-hop QA

The original formulation of RAG is dense, with questions and passages projected into the same vector space by a bi-encoder (Lewis et al., 2020; Karpukhin et al., 2020). The classic lexical alternative, BM25, ranks by weighted term overlap (Robertson and Zaragoza, 2009) and remains in use as a base retriever in recent multi-hop QA work (Trivedi et al., 2023; Jeong et al., 2024). In both cases, the premise is the same. The passage with the answer resembles the question and can therefore be found in a single round of search.

Multi-hop QA was built against it. HotpotQA introduced questions that require combining multiple Wikipedia documents and annotated the *supporting facts* that ground each answer (Yang et al., 2018). However, the fact that a question requires several documents does not mean that the model actually connects them. Trivedi et al. (2020) show that models can reach the answer using only part of the annotated evidence. The authors call this *disconnected reasoning* and propose a way to measure and reduce that behavior. The two benchmarks of this study treat this shortcut in different ways. 2WikiMultiHopQA generates a good share of its questions from structured Wikidata chains, with the evidence paths annotated (Ho et al., 2020). MuSiQue composes multi-hop questions bottom-up, from verified single-hop questions, filtering out compositions that a model would solve without the intermediate hop (Trivedi et al., 2022). These designs produce sets in which a single document is rarely enough, although this happens to different degrees, as Section 6.2 takes up. It is this property that this study exploits by measuring retrieval over corpora built from the supporting documents of these two benchmarks.

For a single-round retriever, the difficulty is structural, and the link that has to be resolved first has a current name in the literature, the *bridge entity*. Until it is identified, the passage that closes the chain shares no surface with the question. One line of work attacks the problem by keeping the index flat and iterating the search. IRCot interleaves chain-of-thought steps with new rounds of retrieval, using the partial reasoning as the next query (Trivedi et al., 2023), and Adaptive-RAG learns to route each question between no round, one round or several, according to the estimated complexity (Jeong et al., 2024).

In both cases, the index remains flat, and the evidence chain is built at query time. When the search is iterated, this requires new calls to the model.

Retrieval, on its own, does not settle the problem either, because the behavior of the reader introduces variance of its own. Language models use the context unevenly according to the position of the relevant evidence. Accuracy falls when it occupies the middle positions and rises when it sits at the ends of the window, a U-shaped profile that the authors observe in sufficiently large models, including those announced as long-context (Liu et al., 2024). They also answer, often, from parametric memory. Mallen et al. (2023) show that, for popular entities, the memory of the model matches or exceeds retrieval, which in large models can even get in the way, and Lewis et al. (2021) document substantial overlap between training and test in the classic open-domain QA datasets, inflating apparent performance. A system with retrieval can receive credit for an answer that was already in the memory of the model. For this reason, the design includes a closed-book control, in which the reader answers without receiving documents. The results are compared to that control and to the dense baseline (Section 3.1).

2.2 Graph-Based RAG Frameworks

Graph-based RAG covers methods in which the retrieval index is structured as a graph, usually created from the extraction of entities and relations by an LLM. The query can then traverse explicit connections instead of relying only on similarity to the question. The name *GraphRAG* is reserved for the specific Microsoft method, and the broad term designates the area.

HippoRAG builds an open knowledge graph with triples extracted with OpenIE and ranks passages with Personalized PageRank (PPR) from the entities of the question (Gutiérrez et al., 2024). In this study, HippoRAG 2 was used, which integrates the passages into the graph itself and refines the selection of seed nodes (Gutiérrez et al., 2025). LightRAG indexes entities and relations with text descriptions alongside dual-level retrieval keys and combines local graph traversal with vector search (Guo et al., 2025).

Microsoft GraphRAG and Cogniee organize the index in another way. Microsoft GraphRAG partitions the graph into hierarchical communities and generates one summary for each. At query time, it operates with local and global search (Microsoft, 2026). Cogniee builds a typed entity graph, which can be guided by an ontology schema. In retrieval, it combines semantic search with context extracted from the graph (Topoteretes, 2026). Although they build and query their indexes in different ways, in the four frameworks retrieval remains coupled to generation in a pipeline of their own.

The published evaluations of these frameworks let the answer layer vary. LightRAG is compared to baselines by the *win rate* assigned by an LLM judge, but each side generates the answers with its own pipeline. What stays constant is the judge, not the answering pipeline (Guo et al., 2025). Microsoft GraphRAG and Cogniee present results in their own materials, without peer review. In those cases, each framework retrieves the evidence with its default components and answers with its own pipeline. The reported delta therefore belongs to the whole package, not only to the substrate.

Earlier work has already controlled the answering stage, although with different scopes. In the HippoRAG 2 evaluation, the baselines that return passages are tested with the same reader, Llama-3.3-70B-Instruct, applied to the first five, and they share the extractor and the dense retriever (Gutiérrez et al., 2025). LightRAG and Microsoft

GraphRAG, on the other hand, continue to answer with their own pipelines. The common prompt is applied only afterward, to standardize the format of the answers, without controlling the reader.

Zhou et al. (2025) extend that control by reimplementing twelve graph-based RAG methods in a single unified framework, with the same LLM, the same embedding model and the same k , on benchmarks that include MuSiQue. Even so, the comparison covers only one answering regime. In a more restricted design, Houimli et al. (2025) compare four retrieval strategies over a fixed biochemical KG, but within a single system and domain.

2.3 Evaluation of RAG Systems

Evaluating a RAG system requires measuring the retrieval of the evidence and the correctness of the final answer. On both datasets, $\text{recall}@k$ compares the retrieved documents to the annotated supporting documents. The correctness of the answer is less direct. Lexical metrics inherited from extractive QA, such as *exact match* and token F1, assume short and canonical answers, while LLM-based readers can correctly phrase the same answer in another way. Kamalloo et al. (2023) show that many answers rejected by *exact match* are accepted by human evaluators and that this difference can change the ranking among systems (Section 4.2).

Using an LLM as a judge has empirical support. Zheng et al. (2023) report agreement between strong LLM judges and human evaluators comparable to the agreement among the humans themselves. ARES and RAGAs separate the evaluation into context relevance, faithfulness and answer relevance (Saad-Falcon et al., 2024; Es et al., 2024). ARES uses a small set of human annotations, while RAGAs computes its metrics without reference human annotations. Imtiyaz et al. (2025) present a review of the evaluation frameworks available up to 2025. Even so, the evaluator can favor a text that it produced itself, above all when it can recognize the text as its own (Panickssery et al., 2024). For this reason, this study measures the agreement between two separate passes of the judge and compares its labels with human evaluation in two protocols, using Cohen’s κ (Cohen, 1960) (Section 4.3).

Validating the judge supports the scoring of the answers, but it is not enough to conclude that two systems differ. Since the systems are evaluated on the same questions, the comparison preserves pairing by item, as recommended for NLP experiments (Dror et al., 2018). The exact McNemar test examines the discordant pairs to check whether the results systematically favor one of the systems (McNemar, 1947). Since the evaluation brings together several comparisons, the p values receive Holm correction within each family of tests (Holm, 1979).

2.4 Attribution and Abstention

A correct answer does not necessarily follow from the retrieved evidence. In the AIS framing, a statement is attributed only when the source provided supports it (Rashkin et al., 2023). This distinction matters in RAG because, faced with conflicting evidence, the model can preserve the answer from parametric memory instead of following the context (Longpre et al., 2021; Wu et al., 2024). The presence of the supporting documents in the context delivered to the reader therefore does not demonstrate that the answer came from them. This study measures the utility of the context through accuracy, but

it does not verify the attribution of each answer.

The amount of context makes that relation even less direct. Random noise can raise accuracy, while increasing the number of documents or the presence of distractors on the same topic can harm reasoning (Cuconasu et al., 2024; Shi et al., 2023). The position of the evidence in the context window also affects its use (Liu et al., 2024). A larger context can thus provide useful evidence and at the same time make reading harder. This ambiguity guides the context-budget audit of Section 3.4.

When the retrieved evidence is insufficient, the system can answer anyway or abstain. In selective prediction, abstention can reduce the risk of error, but it also lowers the proportion of questions answered (El-Yaniv and Wiener, 2010). In QA, SQuAD 2.0 made that decision explicit by including unanswerable questions in the evaluation (Rajpurkar et al., 2018). Kamath et al. (2020) show that it also matters when the system operates outside the training domain. In this study, refusals are recorded separately from incorrect answers.

2.5 Synthesis and Research Gap

In summary, Section 2.1 showed why multi-hop QA requires retrieving and combining evidence distributed across documents. Section 2.2 presented the four frameworks and showed that their evaluations measure the whole package or control only part of the path between retrieval and answer. Section 2.3 grounded the separate evaluation of retrieval and of the answer, including the use of an LLM judge and of paired tests. Finally, Section 2.4 showed why accuracy has to be interpreted in the light of the context delivered and of the decision to answer.

The comparison by Zhou et al. (2025) is the one that comes closest to the control required, but it includes neither HippoRAG 2 nor Cogneer and covers only one answering regime. In the literature consulted, no comparison was found of LightRAG, Microsoft GraphRAG, HippoRAG 2 and Cogneer that evaluates the same indexes both with a common reader and with the native pipeline of each framework. The next chapter presents the two-arm design built to carry out that comparison.

3 Methodology

Figure 1, presented below, follows the experiment from the selection of the data to the statistical evaluation in numbered phases. The output of each phase serves as the input to the next. First, the benchmark records are normalized. The paragraphs they carry form the canonical corpus used in the construction of each index. The same experiment store runs through preprocessing, modeling and evaluation.

3.1 Research Design: the Two-Arm Protocol

When a RAG system gets a question wrong, the cause can lie in different stages:

1. Retrieval: the system selected incorrect documents.
2. Context assembly: the documents were serialized incorrectly.
3. Reader: the prompt and the model produced the incorrect answer.

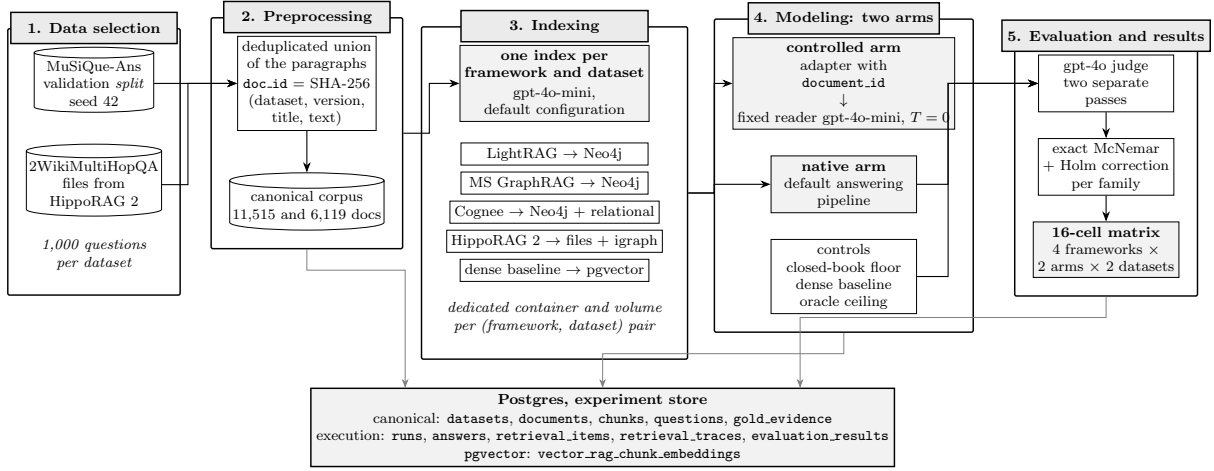


Figure 1: End-to-end methodology.

End-to-end evaluation shows the final result, but it does not reveal in which stage the error occurred, nor where the graph structure brought gains or losses. To deal with this question, the experiment adopts two arms in order to compare retrieval under common conditions with the performance of the complete pipeline. Each framework travels the path from question to answer once in each experimental arm, *native* and *controlled*. Figure 2 illustrates the paths taken by each one.

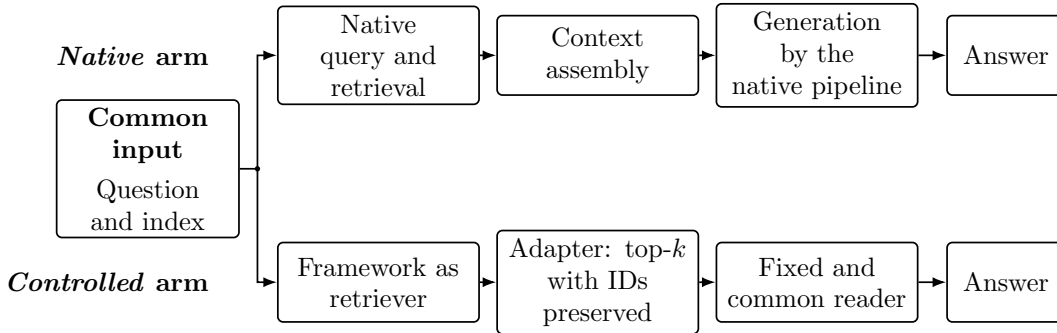


Figure 2: Paths taken by the native and controlled arms.

Both receive the same question and reuse the same index, although they may retrieve different documents. That variation does not occur in HippoRAG 2, whose two arms start from the same five retrieved documents, a condition that allows the reading stage to be compared with the evidence held constant in Section 6.2. In the *native* arm, the framework runs its own pipeline, just as in its standard use. In the *controlled* arm, in turn, the framework acts only as a retriever. An adapter preserves the identification of the framework’s top-ranked documents and supplies them to a fixed reader. That reader, which is common to all the experimental cells, produces the answer from the documents received. The term *controlled* designates the answering stage. Reader, prompt, model, temperature and serialization are identical across the cells of that arm. The retrieval mechanism, the representation returned, the number of documents and the depth of the candidate pool continue to vary, and Section 3.4 quantifies that variation.

The two arms allow the frameworks to be compared under distinct conditions, but that comparison, on its own, is not enough to interpret their results. For this reason, each dataset includes three reference points: the *closed-book* control, the *dense baseline* and the *oracle ceiling*. The *closed-book* control establishes the floor by showing what the reader answers without receiving any document. It thus measures what the reader draws from parametric memory and prevents the evaluation from crediting retrieval with what the model already knew (Mallen et al., 2023). The *dense baseline* retrieves the five closest documents by exact search over embeddings of the OpenAI `text-embedding-3-small` model and generates the answer with the same fixed reader. Finally, the *oracle ceiling* delivers the gold documents directly to the reader, separating retrieval error from reading error. The floor and ceiling labels hold in the aggregate. Per question they do not bound the cells, because a framework gets items right that the oracle gets wrong and gets items wrong that the closed-book control gets right.

3.2 Datasets and Construction of the Corpora

The study uses two multi-hop benchmarks with annotated supporting documents: MuSiQue-Ans and 2WikiMultiHopQA (Trivedi et al., 2022; Ho et al., 2020). Each dataset contributes 1,000 questions. Since the MuSiQue test split does not release the answers, the questions were selected from the validation split. For 2Wiki, the same files from the HippoRAG 2 evaluation were used, preserving comparability with its results (Gutiérrez et al., 2025). In both cases, retrieval takes place over the complete corpus. The annotated supporting documents are treated as gold documents and serve as the reference for computing `recall@5`.¹

The benchmarks differ in the size of the corpora, in the depth of the chains and in the composition of the questions. For this reason, their results are interpreted separately and, when necessary, by question type, as discussed in Section 6.2. The sizes of the corpora and the composition of the questions are presented in Appendix B.

3.3 Frameworks and Configuration Parity

In this study, the frameworks were run in their default configuration, with no re-tuning and no local optimization. This choice represents the conditions found by those who adopt these systems. The exceptions are the ones the protocol requires, and there are four of them: the language model and the embedding model, standardized and recorded in Table 1, the temperature fixed at zero, which also reaches the internal generation of each framework and not only the reader, and the retrieval parameters of Section 3.4. Table 1 records the versions and model snapshots held fixed during the experiment.

To maintain parity among the frameworks, all of them used gpt-4o-mini as the internal model for indexing and generation. Each index went through integrity verifications compatible with its backend. Since the frameworks have different architectures, the scope of the verifications also varied. The details of the verifications can be found in Section 5.2, while the resulting sizes are detailed in Table A.1 of Appendix A.

¹For each question, let G be the set of gold documents and R_5 the set of the first five retrieved documents. `recall@5` is computed as $|G \cap R_5|/|G|$. Thus, if two of the three gold documents appear among the first five results, `recall@5` is 2/3.

Table 1: Component Specification

Component	Version or snapshot	Role
LightRAG	1.5.0	framework
Microsoft GraphRAG	3.1.0	framework
HippoRAG 2	2.0.0a3	framework
Cognee	1.1.2	framework
gpt-4o-mini	gpt-4o-mini-2024-07-18	indexing, generation and reading
gpt-4o	gpt-4o-2024-08-06	judge
Embeddings	text-embedding-3-small	all, including the dense baseline

3.4 Context-Budget Audit

The adapter contract requires every returned item to preserve the identification of the source document, but it does not determine a fixed number of documents. The nominal retrieval parameter is not comparable across the implementations, because each one applies it to a different unit. In the two frameworks that deliver the most distinct documents, the requested value was not 5 either. In LightRAG it was 40, which is the default of the library itself and applies to the entity and relation seeds, while the `chunk_top_k` of 20 in the same library bounds the text chunks that enter the context, and it is this second parameter that explains the 20 documents observed. In Microsoft GraphRAG it was 20, a cut by the harness on the context it assembles. Table 2 presents the unit used by each framework and the number of documents that actually reached the reader.

Table 2: Retrieval unit of each framework, and distinct documents covered by the items delivered to the reader, per dataset (controlled arm). The reader receives the retrieved items, without deduplication, which exists only in the metric.

Framework	Unit of <code>top_k</code>	MuSiQue	2Wiki
Cognee	graph elements (single blob)	2.76	2.46
Dense baseline	chunks, deduplicated by document	4.92	4.82
HippoRAG 2	passages after PPR	5.00	5.00
Microsoft GraphRAG	graph elements (local search)	16.57	12.95
LightRAG	entities/relations (<code>chunk_top_k</code> 20)	20.00	20.00

The amount of context that reaches the reader varies by a factor of 7.2 on MuSiQue and 8.1 on 2Wiki, and the volume is only the part that the table measures. The cells of the controlled arm also differ in which documents arrive, in what order and with how much redundancy.

The imbalance matters, but these results do not say in which direction it acts. Section 6.3 examines what the literature supports in each direction.

4 Design Specification

This chapter specifies how the measurement defined in the Methodology becomes executable and comparable, through the design decisions of the harness and of the instrument used to evaluate retrieval and answers.

4.1 Harness Architecture and Adapter Contract

The harness organizes the runs on the basis of three parameters: dataset, framework and arm. Each combination goes through the same flow. In this way, the inclusion of 2WikiMultiHopQA added new cells to the matrix without requiring the creation of new components. To avoid conflicts among the dependencies of the frameworks, each one runs in an isolated environment and communicates with the harness through a separate process. The details of the environments and of the containers used are presented in Section 5.1.

In the *controlled* arm, the adapter standardizes the result of retrieval. Each item contains the retrieved text and the identifier of the source document (`document_id`). That identifier makes it possible to trace the origin of the passages and to compute the retrieval metrics without depending on the internal representation of each framework. The origin of the identifier varies with the system. In the dense baseline, it comes from the chunk. In HippoRAG 2, it comes from the passage map of the runner. In LightRAG, it comes from the prefix of the chunk identifier. In Microsoft GraphRAG, it comes from the text units associated with the entities and relations included in the context. In Cogneer, it comes from the pattern exposed by the output blob, following the order of first appearance. The metric uses the corpus document that each element points to, not the graph element itself. For the retrieval computation, the list is deduplicated preserving the order of the ranking. The reader receives only the text of the items, in the order in which the framework returns them and without deduplication. In this way, two items can point to the same document.

The fixed reader uses the same versioned prompt in every cell of the controlled arm and in the two controls that receive context: the *dense baseline* and the *oracle ceiling*. The prompt instructs the model to use the retrieved passages when they are useful. If they are not enough, it allows resorting to the knowledge of the model and requests an answer instead of a refusal. It does not require a structured output schema. The full text of the prompt is reproduced in Appendix B. The native outputs vary in size and style, with means between 26 characters in HippoRAG 2 and approximately 1,200 in Microsoft GraphRAG. This difference in format contributes to the choice of the scoring instrument presented in the next subsection.

4.2 Answer Scoring: a Two-Pass LLM Judge

As discussed in Section 2.3, lexical metrics such as token F1 compare the answer to the ground truth by word overlap and can penalize correct formulations that use different terms. To reduce that dependence on format, an LLM judge was adopted (Zheng et al., 2023).

The judge uses `gpt-4o`, with temperature 0 and a versioned prompt (Appendix B), and receives the question, the ground truth with its aliases and the answer produced. Figure 3 summarizes that path, from the common input and the generation of the answers in the two arms to the judgment in two separate passes.

In each pass, the judge classifies the answer as *correct*, *partial*, *incorrect*, *refusal* or *gold suspect*. The last label indicates a possible problem in the ground truth and removes the item from the denominator. Table B.3 of Appendix B presents the criteria and an example of each case.

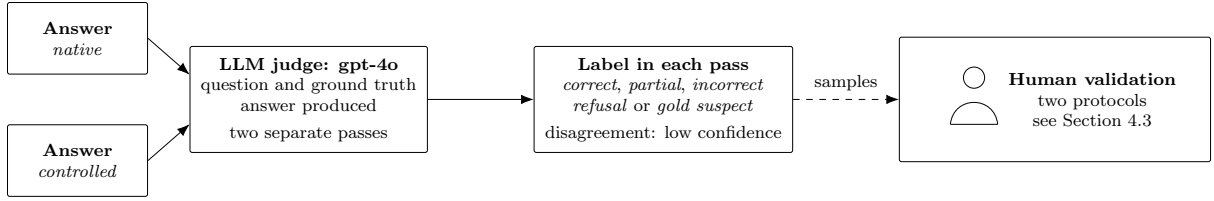


Figure 3: Answer scoring flow and human validation of the judge.

The judgment considers what the answer asserts, not its length. A long answer can be considered correct, while merely mentioning the expected entity is not enough if the entity is not presented as the answer.

Each item is judged twice, with the order of the *few-shot* examples inverted between the passes. Inspired by Zheng et al. (2023), this inversion is used as a stability check. The label of the first pass determines the score of the item. The second pass is used only to measure stability. If the two disagree, the item is marked as low confidence, without any change to the score. Strict accuracy corresponds to the proportion of items labeled *correct* in the scoreable base, which excludes the *gold suspect* ones. The refusal and incorrect-answer rates are computed from the same labels.

4.3 Human Validation of the Judge

Since the LLM judge is part of the evaluation instrument, it was submitted to two human validation protocols, both conducted by the author. In the primary protocol, a stratified sample of the answers was analyzed, with access to the labels assigned by the judge. The supplementary protocol concentrated on the most fragile cases and kept those labels hidden during annotation. The identity of the framework and of the experimental arm, however, remained visible. Since the samples follow different criteria, their results are reported separately. Table B.4 of Appendix B brings together the results of each protocol. The limitations are discussed in Section 6.3.

5 Implementation

The implementation preserves the isolation between cells and makes the construction of the indexes repeatable, without creating a new experimental condition.

5.1 Environment and Tools

The evaluation harness was implemented as a Python package compatible with version 3.11 or above. Its dependencies were resolved and pinned by a lock file. Each framework was installed in a dedicated virtual environment, according to the version indicated in Section 3.3.

Storage follows the same principle of isolation. The containers are managed by Podman, a container management tool, and run without administrator privileges. A PostgreSQL instance with the `pgvector` extension stores the experiment records and the *embeddings* of the dense *baseline*. Each combination of framework and dataset that uses Neo4j has its own Community Edition instance, with a dedicated volume and port. Cognee also uses a separate relational instance, required by its architecture. HippoRAG 2

does not need a container, since its index is stored in local files and traversed with `python-igraph`.

5.2 Preparation of the Corpora and Index Construction

The two benchmarks were ingested into the canonical format and persisted in the store, applying the sampling and deduplication decisions of Section 3.2. Before the indexes were built, the harness split the documents that exceeded 1,200 characters. Each index was then built once per framework and dataset, in the respective isolated environment. In the frameworks with incremental ingestion, the construction runs in batches with a persisted checkpoint, so that an interrupted construction continues from where it stopped instead of starting over. HippoRAG 2 indexes the whole corpus in a single pass.

Over the same MuSiQue corpus, Microsoft GraphRAG produced 51,115 nodes and Cognee 92,573. The difference comes from the extraction decisions of each framework, not from local configuration, and the complete counts, by framework and dataset, are in Table A.1 of Appendix A.

The integrity check looks for duplication capable of inflating retrieval without leaving a visible signal. It was adapted to what each backend allows, and it did not cover every index in the same way. The two Cognee indexes were reconciled against the canonical document record, and the number of distinct identifiers matched exactly, with zero duplications. In the LightRAG and Microsoft GraphRAG indexes on 2WikiMultiHopQA the check was a different one, of duplication internal to the graph. It also came out at zero. Since the scope differed between backends, these results hold for the cases checked and do not rule out duplication uniformly across the matrix.

6 Evaluation

6.1 Main Results

The results cover the sixteen cells and the three controls per dataset defined in Chapter 3. Retrieval is examined first, because the answer score does not show whether the reader received the annotated evidence. The metrics `recall@5`, `all_gold@5` and `recall@pool` describe that stage, while strict accuracy (Section 4.2) measures the answer and serves as the basis for the paired tests. Since pool depth varies across the methods, the direct comparison between them uses `recall@5`. `recall@pool` is interpreted only within the finite list persisted in each run. Appendix B details the denominators and the notation.

Retrieval failure modes and what the run makes visible

Table 3 brings together the budget delivered by each adapter, the evidence retrieved at the cut-off and in the persisted pool and, on the right, strict accuracy with the delta against the dense baseline. This arrangement makes it possible to check whether the loss appears in retrieval or in the answering stage.

The table separates the methods into two groups, and the division is one of observability before it is one of behavior. In LightRAG and in Microsoft GraphRAG, the pool extends beyond the cut-off. The difference between `recall@5` and `recall@pool` makes visible the evidence that is retrieved but placed outside the first five results. On MuSiQue,

Table 3: Controlled arm: budget delivered, retrieval and answer per method. “docs/q” is the mean number of documents the adapter returns per question. The retrieval metrics are computed with $n = 1,000$. Strict accuracy uses the scoreable base of each cell. The delta against the dense baseline comes from the paired test over the intersection of the scoreable items of the two cells. Significance by exact McNemar with Holm correction: *** $p < 0.001$, * $p < 0.05$, no mark = not significant.

Framework	docs/q	recall@5	recall@pool	all_gold@5	strict	vs. dense
<i>MuSiQue</i>						
Closed-book floor					0.209	
Dense baseline	4.92	0.553	0.553	0.218	0.463	
HippoRAG 2	5.00	0.603	0.603	0.303	0.502	+3.8*
LightRAG	20.00	0.506	0.704	0.181	0.519	+5.7***
Microsoft GraphRAG	16.57	0.447	0.615	0.167	0.495	+3.2*
Cognee	2.76	0.434	0.434	0.109	0.401	−6.2***
Oracle ceiling					0.770	
<i>2WikiMultiHopQA</i>						
Closed-book floor					0.335	
Dense baseline	4.82	0.704	0.704	0.409	0.678	
HippoRAG 2	5.00	0.864	0.864	0.672	0.801	+12.2***
LightRAG	20.00	0.693	0.777	0.400	0.731	+5.3***
Microsoft GraphRAG	12.95	0.608	0.670	0.298	0.694	+1.6
Cognee	2.46	0.625	0.625	0.266	0.615	−6.3***
Oracle ceiling					0.890	

this appears in LightRAG: the last gold document occupies, on average, position 6.78, just after the cut-off.

In the dense baseline, in HippoRAG 2 and in Cognee, the pool ends at the context delivered to the reader. The equality between **recall@5** and **recall@pool** follows from that construction and does not allow missing coverage to be separated from evidence that could appear after the fifth position. In Cognee, however, the limitation is visible in the context itself: with 2.76 documents per question, no annotated evidence reaches the reader in 160 of the 1,000 MuSiQue questions.

On 2WikiMultiHopQA, the main difference lies in HippoRAG 2. Its **recall@5** reaches 0.864, against 0.704 for the dense baseline. **all_gold@5** is higher on this dataset for the five methods, but the contrast does not isolate the effect of chain depth, since it also incorporates differences in recall within each stratum.

The per-framework effect in the controlled arm

The last column of Table 3 compares each retrieval adapter to the dense baseline under the fixed reader. The counts of discordant pairs and the p values are in Table A.2 of Appendix A.

LightRAG and HippoRAG 2 beat the dense baseline on both datasets, while Cognee falls below it on both. In Microsoft GraphRAG, the difference is detectable only on MuSiQue: 3.2 points, against 1.6 points with no detectable difference on 2WikiMultiHopQA. In Cognee, the deficit cannot be separated from the smaller budget delivered by its adapter.

On MuSiQue, in turn, HippoRAG 2 obtains a **recall@5** of 0.603, while LightRAG reaches a strict accuracy of 0.519, against 0.502 for HippoRAG 2, even with a **recall@5**

of 0.506. Retrieving better is not the same as answering better.

The native arm and the gap between arms

In the native arm, the family of six pairs ($m = 6$) resolves into three levels on both datasets, and what changes is who occupies them. On MuSiQue, LightRAG stands alone at the top and Cognee alone at the bottom, with Microsoft GraphRAG and HippoRAG 2 at the intermediate level, with no detectable difference between them (+0.8 pp, $p = 0.670$). On 2WikiMultiHopQA, HippoRAG 2 occupies the top alone and LightRAG the intermediate level, while Microsoft GraphRAG and Cognee share the bottom, also with no detectable difference (−1.6 pp, $p = 0.461$). The other five comparisons in each dataset show a detectable difference after Holm correction. Switching arms does not change which method comes first in each dataset: LightRAG on MuSiQue and HippoRAG 2 on 2WikiMultiHopQA.

To compare the two modes of execution, each framework is then set against itself in the native and controlled arms, over the same index. Table 4 presents the two accuracies, the paired delta, the p value and the refusal rate.

Table 4: Native arm against its own controlled arm, over the same index. Accuracies over the scoreable base of each cell. Paired deltas over the intersection of the scoreable items of the two cells, and for that reason they can differ from the subtraction of the rounded values. A positive delta means that the native pipeline scores above the controlled reader. Refusal is that of the native arm, over the 1,000 answers of the cell. Exact McNemar with Holm correction in the family of the gap between arms ($m = 4$ per dataset).

Framework	controlled	native	Δ (pp)	Holm p	refusal
<i>MuSiQue</i>					
LightRAG	0.519	0.548	+2.8	0.142	11.9%
Microsoft GraphRAG	0.495	0.474	−2.1	0.168	9.2%
HippoRAG 2	0.502	0.466	−3.5	0.024	15.1%
Cognee	0.401	0.349	−5.2	< 0.001	14.8%
<i>2WikiMultiHopQA</i>					
LightRAG	0.731	0.615	−11.6	< 0.001	21.8%
Microsoft GraphRAG	0.694	0.518	−17.6	< 0.001	26.0%
HippoRAG 2	0.801	0.723	−7.7	< 0.001	16.0%
Cognee	0.615	0.534	−8.0	< 0.001	20.9%

For MuSiQue, the difference between the arms is mixed. LightRAG gains 2.8 points and Microsoft GraphRAG loses 2.1, with no detectable difference in either case. HippoRAG 2 and Cognee show significant drops. On 2WikiMultiHopQA, the four frameworks lose between 7.7 and 17.6 points, with significance in every case. On that dataset, native refusal lies between 16.0% and 26.0%, against 1.6% and 4.1% in the controlled arm.

The arms also distribute the error in different ways. In the controlled arm, it appears above all as an incorrect answer, while in the native arm part of it becomes a refusal. Two cells with similar accuracy can therefore show distinct profiles. The preference between those profiles depends on the relative cost of error and refusal, which was not estimated in this study.

Section 6.2 takes up the difference between the datasets in the light of the question type.

6.2 Patterns that Help Interpret the Results

The aggregates change according to the type of reasoning represented in each benchmark. Two relations help to interpret them. Chain depth is associated with the probability of gathering all the evidence at the cut-off, while the question type affects the possibility of answering correctly without traversing the intended chain. These analyses were not pre-registered and were defined after the inspection of the results. For this reason, the decompositions in this section are descriptive. They condition the matrix by stratum, but they do not formally test differences between them.

Chain depth against the budget

As the number of required gold documents grows, the proportion of questions in which the annotated chain appears in the first five positions falls. The variation is measured by `all_gold@5`. Figure 4 brings together the results by chain depth. Between two and three required documents, the probability of assembling the chain falls by a factor of 6.6 in the dense baseline and 24.1 in Microsoft GraphRAG. The five positions are a measurement convention, not the effective budget, which varies by framework and dataset.

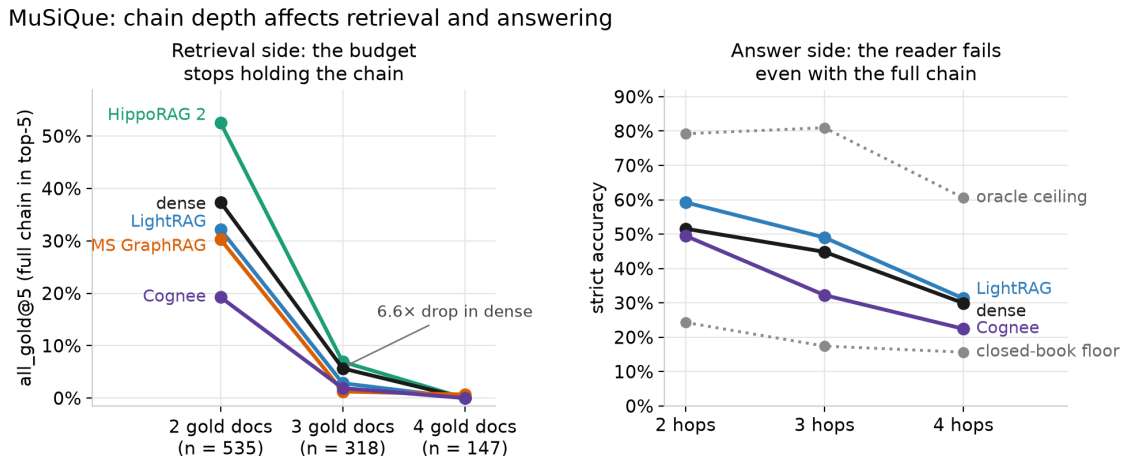


Figure 4: Chain depth on MuSiQue. On the left, `all_gold@5` for the five methods, with $n = 1,000$. On the right, the strict accuracy of LightRAG, of the dense baseline and of Cognee, alongside the closed-book and oracle controls, over the scoreable base of each stratum.

The effect of depth differs between the benchmarks. On MuSiQue, 465 of the 1,000 questions require three or more documents (Section 3.2). 2Wiki is bimodal, and its 235 four-document questions belong to the `bridge_comparison` type. In that stratum, the question literally presents a closed set of candidates in 85% of the cases. The dense baseline reaches 0.821 even with `all_gold@5` close to zero, a pattern consistent with the *disconnected reasoning* formalized by Trivedi et al. (2020). In the DiRe test of Trivedi et al. (2022), 2Wiki obtains 63.4, against 37.8 for MuSiQue-Ans. In that stratum, the closed-book control reaches 0.519, a result also consistent with that reading. On MuSiQue, the questions do not offer an equivalent closed set.

The difference by depth also appears in the reading stage. In the oracle control, which receives the annotated gold documents, strict accuracy goes from 0.792 on two-document questions to 0.605 on four-document ones, a difference of 18.7 points.

Composition by question type

The aggregate comparison between datasets brings together strata of different difficulty. Table A.3 of Appendix A presents the decomposition of 2Wiki by question type. In the **comparison** type, the dense baseline reaches 0.943, against 0.975 for the oracle control, a difference of 3.2 points. The **compositional** stratum, in turn, shows values close to those of the MuSiQue aggregate: 0.194 against 0.209 in the closed-book control and 0.484 against 0.463 in the dense baseline. That proximity offers a point of comparison between the datasets and helps to interpret the aggregate difference of 21.5 points between their dense baselines. Retrieval difficulty also differs, as Table 3 shows, and the design does not estimate the contribution of the two factors separately. In the **compositional** stratum, the deltas against the dense baseline are 11.1 points for LightRAG and 13.1 for Microsoft GraphRAG, against 5.3 and 1.6 points in the aggregate.

The decomposition also helps to interpret the Microsoft GraphRAG result left open in Section 6.1. When the strata are weighted by size, **compositional** contributes 5.4 points to the aggregate, while **comparison** takes away 3.5. The other two strata take away another 0.3, resulting in the 1.6 points observed. This composition is consistent with the dilution hypothesis, without ruling out a degradation of the method in **comparison**. In Cognee, the deltas remain negative in the four question types.

Taken together, these patterns support a descriptive hypothesis. Graph-based pipelines can show larger gains in **compositional**, where the chain has to be discovered, and null or negative gains when the question already names the candidates. Since the context budget is not equalized (Section 3.4), this hypothesis applies to the complete retrieval pipelines. The data do not isolate the effect of graph structure.

The internal comparison with constant context

Composition helps to situate the aggregate difference between the datasets, but it does not show what changes after retrieval. That comparison is possible only in HippoRAG 2, because its two arms receive the same five documents (Section 3.1). The analysis below uses 2WikiMultiHopQA, and Table 5 presents the complete distribution of the verdicts.

Table 5: HippoRAG 2 on 2WikiMultiHopQA: distribution of the verdicts in the two arms over the same five retrieved documents. Proportions over the scoreable base of each cell. The four classes sum to 1 in each row. The differences between the rows are marginal and can diverge from the paired delta of Table 4, computed over the intersection of the scoreable items.

HippoRAG 2, 2Wiki	strict	partial	incorrect	refusal
Controlled arm	0.801	0.034	0.149	0.016
Native arm	0.723	0.044	0.073	0.160

In the marginal proportions of Table 5, strict accuracy falls 7.8 points. Incorrect answers fall 7.6 points, while the *partial* label rises 1.0 point and refusal, 14.4. In the paired comparison over the intersection of the scoreable items, the accuracy delta is 7.7 points and remains significant after Holm correction ($p < 0.001$). This redistribution occurs after retrieval, but the analysis does not estimate the contribution of each component of the reading pipeline. In the other three frameworks, the context also changes between the arms.

6.3 Validation and Threats to Validity

The threats are ordered by what each one puts at risk, from the instrument that produced the numbers to the text of the corpora.

All the results in this chapter pass through an automatic judge, validated by the two protocols of Section 4.3. The primary protocol started from the label of the judge and recorded total agreement, with a κ of 1.000, a value that measures consistency and not independence. The independent evidence comes from the blind supplement of 30 items, with 93.3% agreement and 16 of 16 on the refusals of the native arm. With 16 items, the 95% Wilson interval still admits a true rate of 0.81 at the lower bound, and that is the margin available.

The context budget is what weighs most on the design. The audit of Section 3.4 found unequal budgets under nominal parameters that are not comparable across the implementations, and the inequality, on its own, does not determine the direction of the effect. Additional documents can bring useful evidence or noise that distracts, and the literature reviewed in Section 2.4 supports both readings (Cuconasu et al., 2024; Shi et al., 2023; Liu et al., 2024).

Pool depth was not equalized either, and a probe on the dense baseline shows what that hides. In a single search over twenty chunks, the recall of that baseline goes, in the same probe, from 0.555 at the cut-off of five to 0.735 over the twenty, above the **recall@pool** of 0.704 of LightRAG in the canonical audit. The same method that appears in Table 3 with pool equal to the cut-off therefore has evidence below the fifth position. The probe covers MuSiQue and was not repeated on 2WikiMultiHopQA. Separating coverage from ranking in the methods where the pool ends at the cut-off would require deeper pools, a run that Section 7.2 records.

The interpretation by composition and chain depth was weighed against properties of the corpus text itself. On 2Wiki, the supporting documents have 151.1 tokens on average against 87.0 for the distractors, and the **recall@5** of the long-gold questions rises for the five methods, between 2.8 and 8.6 points. On MuSiQue the two groups are indistinguishable, and the same cut raises only the dense baseline. The association is worth five points in the dense baseline itself, against the fifteen that separate its recall on the two datasets.

7 Conclusion

Based on the questions stated in Section 1, an experimental design was developed and applied in full, represented by: the sixteen-cell matrix (with the three controls of each dataset), the paired tests, and the human validation of the judge.

The first question compared four graph-based RAG frameworks in the two arms, with dense retrieval as the baseline of the controlled arm. Under the fixed reader, what separates the results is the framework, not the category. On both datasets, LightRAG and HippoRAG 2 end above the dense baseline and Cognee about six points below, and only for Microsoft GraphRAG does significance depend on the dataset. Under the native pipelines the top of the ordering changes with the dataset, with LightRAG ahead on MuSiQue and HippoRAG 2 on 2WikiMultiHopQA. None of these deltas isolates the substrate, because of the limit that Section 7.1 states. In the only framework where the documents stay constant between the arms, switching the reading pipeline moves the result, with the share of refusal still to be isolated.

The second question was how much the budget, the composition and abstention help to interpret those differences. Within that same limit, the composition of each dataset supports a descriptive reading of the aggregate difference between the two (even though retrieval difficulty also differs and this design does not separate the two). These decompositions by stratum were defined after the aggregates had been inspected. Abstention separates the arms on 2WikiMultiHopQA, where the native pipelines refuse between one in six and one in four questions, which penalizes them under strict accuracy and shifts the error from the incorrect answer to refusal. HippoRAG 2 documents that shift under constant retrieved context.

None of these answers can be read in the aggregate. Each one required looking separately at the retrieval of the annotated evidence, the correctness of the answer and refusal, and it is that separation, more than any ordering of frameworks, that the two-arm design delivers.

7.1 Limitations

The main limitation is the context budget that was not equalized, from 2.46 to 20.00 documents delivered under nominal parameters that do not define a common unit across the implementations. The direction of the resulting bias is not identifiable from these data, which prevents any causal claim about graph structure. A related hypothesis attributes part of the correct answers to the memory of the reader, and it was not tested. Identifying the source of an answer would require demonstrating, at the same time, that it is correct, that the context does not support it, that it exists in the parametric knowledge of the model and that there is no alternative evidence in the retrieved material. What does exist is the closed-book floor. It is higher on 2WikiMultiHopQA because almost half of its questions name the candidates in the question (Section 3.2). The findings of Longpre et al. (2021) and of Mallen et al. (2023) make the hypothesis plausible as a mechanism, but neither of them makes it a measurement.

This limit restricts the interpretation of the comparisons between frameworks. The results characterize the retrieval adapters and the answering pipelines in the conditions evaluated, without isolating the effect of graph structure.

7.2 Future Work

Each of these limitations defines an experiment. Normalizing the controlled arm to a fixed number of canonical documents would give the reader the same document unit in every cell, and would allow document selection to be separated from the volume delivered. Running the retrievers with deeper pools, with curves at k equal to 5, 10 and 20, would make observable the distinction between coverage and ranking in the methods where today it is not. Verifying attribution item by item would show whether the correct answer derives from the evidence received or from the memory of the model, which requires assertion-based judgment and was not done here. Repeating the design with other readers would indicate whether the orderings and the gap between the arms depend on gpt-4o-mini.

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A Supplementary Statistical Results

This appendix keeps the statistical detail that allows the auditing of the claims of Chapter 6 without being necessary to follow the argument. Each piece was referenced in the body at the point where it supports a reading.

Table A.1: Size of the built indexes, per framework and dataset. The four counts do not measure the same object. LightRAG and Microsoft GraphRAG count entities and relations extracted by LLM, and the nodes of Microsoft GraphRAG do not include the community layer, reported separately. HippoRAG 2 mixes entity nodes with one node per document of the corpus, and its edges include synonymy by embedding similarity. Cognee reports the graph that the retrieval traverses. The dense baseline does not build a graph. The graph of HippoRAG 2 is undirected and inserts part of the relations in both directions, so that it contains parallel edges: its 493,714 edges in MuSiQue cover 319,817 distinct pairs of nodes, and the 304,379 in 2WikiMultiHopQA cover 186,840.

Framework	MuSiQue		2WikiMultiHopQA	
	nodes	edges	nodes	edges
LightRAG	56,161	58,631	29,770	27,677
Microsoft GraphRAG	51,115	61,466	26,589	27,680
HippoRAG 2	112,260	493,714	55,907	304,379
Cognee	92,573	231,517	50,084	116,772

Microsoft GraphRAG adds a community hierarchy, with one generated summary per community: 9,053 communities in seven levels in MuSiQue and 4,222 in five levels in 2WikiMultiHopQA. The corpora have 11,515 and 6,119 documents.

Table A.2: Controlled arm against the dense baseline. Exact McNemar over discordant pairs, Holm correction inside the family ($m = 4$ per dataset). b = questions that only the framework answers correctly, and c = questions that only the dense baseline answers correctly.

Framework	Δ (pp)	MuSiQue			Δ (pp)	2WikiMultiHopQA		
		b	c	Holm p		b	c	Holm p
LightRAG	+5.7	113	57	< 0.001	+5.3	85	32	< 0.001
Microsoft GraphRAG	+3.2	111	79	0.024	+1.6	114	98	0.303
HippoRAG 2	+3.8	107	69	0.010	+12.2	165	43	< 0.001
Cognee	−6.2	58	119	< 0.001	−6.3	42	105	< 0.001

Table A.3: 2WikiMultiHopQA, controlled arm, per question type. The first three columns are the control cells. The last four report the delta of each framework against the dense baseline, in percentage points. Each delta is computed over the intersection of the scoreable items of the two cells, and not by the subtraction of the control columns: in the **compositional** stratum Cogne and HippoRAG 2 have 412 scoreable items against 413 for the others. The last line repeats the aggregate for reference.

Type	floor	dense	oracle	LightRAG	MS GraphRAG	HippoRAG 2	Cogne
compositional (413)	0.194	0.484	0.818	+11.1	+13.1	+26.2	−7.3
inference (108)	0.102	0.509	0.833	+6.5	−1.9	+14.8	−11.1
comparison (244)	0.500	0.943	0.975	+1.2	−14.3	0.0	−3.3
bridge_comparison (235)	0.519	0.821	0.953	−1.3	−0.4	−0.9	−5.5
aggregate (1,000)	0.335	0.678	0.890	+5.3	+1.6	+12.2	−6.3

B Evaluation Instrument

The records of the benchmarks and the evaluation instrument that Chapters 3 and 4 summarize are kept here. The body of the text gives what is enough to read the results.

Sampling bases and notation. In MuSiQue, the answer metrics use the canonical scoreable base of 992 items, after excluding eight labeled *gold_suspect*, dropping to 991 in the cells with one additional non-scoreable item. In 2WikiMultiHopQA, this base is of 1,000 items, with 999 in two cells. The paired tests always run over the intersection of the two cells compared, and for this reason a delta can rest on a base slightly smaller than that of the printed accuracies. The refusal rate is the exception. As the canonical source records it, it has as denominator the 1,000 answers of the cell, and not the scoreable base. The retrieval analyses use the 1,000 questions in both datasets. The deltas in prose come from the cross-dataset consolidated file and are not recalculated from the rounded values of the tables. As for notation, metric values on a scale of 0 to 1 take a decimal point, in the manner of the sources, and counts, document means and percentage points take the decimal point too, with the comma reserved for the thousands separator.

Table B.1: The two canonical corpora. Tokens counted with the `cl100k_base` encoder over the text of each document. The totals are exact sums, not multiplied means. The last column counts the documents that exceed the segmentation threshold of the harness, 1,200 characters.

	Questions	Docs	Tokens	Tokens/doc	Segmented
MuSiQue-Ans	1,000	11,515	1,324,087	115.0	373 (3.2%)
2WikiMultiHopQA	1,000	6,119	682,480	111.5	439 (7.2%)

Table B.2: Composition of the 1,000 questions of each dataset. The two benchmarks classify by different criteria, and for this reason the columns do not align: MuSiQue labels by the chain depth and 2WikiMultiHopQA by the type of reasoning. The column on the right translates both into the same unit, the number of gold documents that the question requires.

Dataset	Stratum	Questions	Gold docs
MuSiQue-Ans	2 hops	535	2
	3 hops	318	3
	4 hops	147	4
2WikiMultiHopQA	compositional	413	2
	comparison	244	2
	bridge_comparison	235	4
	inference	108	2

Table B.3: The five labels of the judge. The examples are real items of the LightRAG cell in the controlled arm of MuSiQue, with the predictions abbreviated. The `gold_suspect` label does not receive an example because it marks a defect of the ground truth, not a behavior of the system.

Label	Criterion	Example: ground truth $\rightarrow$ prediction
<code>correct</code>	asserts the ground truth or a clear equivalent	<i>built in the 15th century</i> $\rightarrow$ “constructed between 1400 and 1450”
<code>partial</code>	chain partially resolved, or hesitant answer that includes the correct one	<i>Chad Basin</i> $\rightarrow$ “died in the geological area of Chad”
<code>incorrect</code>	commits to a wrong answer	<i>New England Patriots</i> $\rightarrow$ “playing for the New York Jets”
<code>refusal</code>	refuses or declares that there is no information	<i>January 20, 2017</i> $\rightarrow$ “does not have a specific start date mentioned”
<code>gold_suspect</code>	ground truth apparently defective, excluded from the denominator	

Table B.4: The two protocols of human validation of the judge. The results are reported separately, and not summed, because the samples have deliberately different distributions.

	Primary protocol	Supplementary protocol
Items	120, 60 per dataset	30, 15 per dataset
Sampling	stratified with a fixed seed by the nine judged cells of each dataset, with natural prevalence of labels	deliberately unbalanced, over-sampling native refusals, cases in which the annotated evidence arrived incomplete or did not arrive at the reader, and inter-pass disagreement items
Format	verification and correction, starting from the label of the judge	labeling blind to the label of the judge, with the human column empty. The identity of the framework and of the arm remained visible
Result	agreement 100%, Cohen’s κ (Cohen, 1960) = 1.000 in the five classes and in the strict binary form	agreement 93.3% (28 of 30), with 16 of 16 in the native refusals
Limit	the anchoring in the label of the judge tends to inflate the measured agreement	the sample is unbalanced, and for this reason κ is not computed and global accuracy is not estimated

Prompts

The prompts are reproduced in English, as they run in the harness. The braces mark the fields filled in at run time. The native arm does not appear here, because each framework answers with the prompt of its own library.

Fixed reader. Used in all the cells of the controlled arm and in the controls with context, including the dense baseline and the oracle ceiling. Each retrieved chunk enters on one line, preceded by its identifier. When the retrieval returns an empty list, the block is replaced by `No retrieved chunks`.

Question:
{question}

Retrieved chunks:
{chunk_id}: {chunk_text}
...

Instructions: answer the question. Use the retrieved chunks above when they help, and otherwise rely on your own knowledge; give your best answer rather than declining.

Closed-book control. The parametric floor does not receive context, and for this reason it also does not receive an anchoring instruction.

Question:
{question}

Instructions: answer the question directly from your own knowledge. Do not ask for additional context. If you do not know the answer, say that you do not know instead of guessing.

Judge. Each item is judged in two independent passes. The block below is the system message of pass A. In pass B, the order of the six examples is inverted and the reminder indicated at the end is added, which gives the measure of inter-pass agreement. The output is forced to the schema `{label, matched_gold, rationale}`, with `label` restricted to the five labels of Table B.3.

You grade SHORT-ANSWER multi-hop QA (MuSiQue). You are given a question, the gold answer plus aliases, and a model prediction. Decide whether the prediction ASSERTS the gold answer (or a clear equivalent: synonym, abbreviation, alternate name, different date/number format, same entity), IGNORING verbosity, markdown, restating the question, or extra explanation.

Labels:

- correct: the prediction asserts the gold answer (or a valid equivalent) as THE answer.
- partial: a multi-hop chain partly resolved - right intermediate but wrong/incomplete final; or it hedges between candidates including the right one; or a correct-but-less-specific form.
- incorrect: it commits to a WRONG final answer (wrong entity/value, or answered the wrong sub-question).
- refusal: it declines / says not enough information / denies the premise / is empty.

- gold_suspect: the gold/aliases look wrong, ambiguous or broken, so the item cannot be judged fairly.

Decision rules:

- Verbosity/markdown/preambles NEVER penalize.
- Require ASSERTION, not mere mention: if the gold appears only as context or as one option among several, it is NOT correct.
- When unsure between partial and correct, choose partial.
- Output ONLY the structured JSON (label, matched_gold, rationale<=1 sentence).

[pass B, added here]

Reminder: a verbose answer whose ASSERTED answer is the gold is 'correct'; an answer that only mentions the gold in passing or as one option is NOT 'correct'.

Examples:

Q: Who sings Home Alone Tonight with the singer of Light It Up?

GOLD: Karen Fairchild

ALIASES: []

PREDICTION: **Luke Bryan sings "Home Alone Tonight" with Karen Fairchild.**

JSON: {"label": "correct", "matched_gold": "Karen Fairchild",
"rationale": "asserts the gold; only wrapped in markdown/sentence"}

Q: When was the painter of Femme nue couchee born?

GOLD: 10 June 1819

ALIASES: []

PREDICTION: Gustave Courbet, the painter, was born on June 10, 1819.

JSON: {"label": "correct", "matched_gold": "10 June 1819",
"rationale": "same date, different format"}

Q: What team is the highest goal scorer in the EPL a member of?

GOLD: Egypt national football team

ALIASES: []

PREDICTION: The highest scorer is Mohamed Salah, who is a member of Liverpool FC.

JSON: {"label": "partial", "matched_gold": "Egypt national football team",
"rationale": "right player, gives the club not the requested national team"}

Q: Who was honored with the award Dhondo Keshav Karve received prior to becoming president of India?

GOLD: A.P.J. Abdul Kalam

ALIASES: ["Abdul Kalam", "Kalam"]

PREDICTION: The award is the **Bharat Ratna**.

JSON: {"label": "incorrect", "matched_gold": null,
"rationale": "answered the award, not the person honored with it"}

Q: When did the town WIZE is licensed in become capitol of the state where Ward Township is located?

GOLD: 1839

ALIASES: []

PREDICTION: Springfield has never been the capital of Ohio; the capital is Columbus. There is no relevant answer.

JSON: {"label": "refusal", "matched_gold": null,
"rationale": "denies the premise, gives no answer"}

Q: What is the meaning of the majority religion's word in the Arabic dictionary, in the area that became India?

GOLD: the country of India
ALIASES: ["India", "Hindustan"]
PREDICTION: The word is "Islam", which in Arabic means "submission" or "surrender".
JSON: {"label": "gold_suspect", "matched_gold": null,
 "rationale": "gold 'the country of India' does not answer a 'meaning of the word' question"}

The user message carries the item to be judged, in the same format as the examples and without the JSON line.

Q: {question}
GOLD: {gold}
ALIASES: {aliases}
PREDICTION: {prediction}